



Ludovico Battista

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Contact Information

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Positions

'26 - Feb '27 **Post-doc**, *Max Planck Institute for Mathematics*, Bonn, Germany.
'23 - '26 **Post-doc**, *FBK - Fondazione Bruno Kessler*, FM - Formal Methods, Trento.
'22 - '23 **Post-doc**, *Alma Mater Studiorum - University of Bologna*, Department of Mathematics, Bologna.

Education

'18-'22 **Ph.D. in Pure Mathematics**, *Università di Pisa*, Pisa, 08/04/2022, cum laude.
PhD Thesis: "Hyperbolic 4-manifolds, perfect circle-valued Morse functions and infinitesimal rigidity"
Supervisor: Prof. Bruno Martelli.
'16-'18 **Graduate Student in Pure Mathematics**, *Università di Pisa*, Pisa, Master Degree in Mathematics, 110/110 cum laude, 26/10/2018.
Master Thesis: "Principal congruence Link complements"
Advisor: Prof. Bruno Martelli.
Reward: the thesis was awarded a prize by *Istituto Nazionale di Alta Matematica*.
'13-'16 **Bachelor Student in Pure Mathematics**, *Università di Pisa*, Pisa, Bachelor Degree in Mathematics, 110/110 cum laude, 13/05/2016.
Bachelor Dissertation: "Crescita di gruppi: un gruppo con crescita intermedia" (Group growth: a group with intermediate growth).
Advisor: Prof. Roberto Frigerio.

Preprints

- 1 **Can You Hear the Shape of a Hyperbolic Surface? Now for Real**, *j.w. with J. Souto*, Preprint, 2026.
ArXiv

Publications

- 1 **Hyperbolic 4-manifolds with perfect circle-valued Morse functions**, *j.w. with Martelli, B.*, Transactions of the American Mathematical Society 375.04: 2597-2625, 2022.
arXiv Journal
- 2 **Infinitesimal Rigidity for Cubulated Manifolds**, Geom Dedicata 217, 33, 2023.
arXiv Journal
- 3 **Dodecahedral L -spaces and hyperbolic 4-manifolds**, *j.w. with Leonardo Ferrari and Diego Santoro*, Communications in Analysis and Geometry, 2022.
arXiv Journal
- 4 **Bounded Cohomology Classes of Exact Forms**, *j.w. with S. Francaviglia, M. Moraschini, F. Sarti, A. Savini*, Proceedings of the American Mathematical Society, 2022.
arXiv Journal
- 5 **SMT-Based Stability Verification of an Industrial Switched PI Control Systems**, *j.w. with S. Basagiannis, A. Becchi, A. Cimatti, G. Giantamidis, S. Mover, A. Tacchella, S. Tonetta, V. A. Tsachouridis*, 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks, DSN Workshop VERDI, 2023.
Journal
- 6 **Stability Verification of an Industrial Switched PI Control Systems**, *j.w. with S. Basagiannis, A. Becchi, A. Cimatti, G. Giantamidis, S. Mover, A. Tacchella, S. Tonetta, V. A. Tsachouridis*, Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems, 2024.
EasyChair
- 7 **Formal Verification of Stability for Parametric affine Switched Systems**, *j.w. with S. Tonetta*, 8th IFAC Conference on Analysis and Design of Hybrid Systems ADHS 2024, 2024.
ScienceDirect
- 8 **Deriving Liveness Properties of Hybrid Systems from Reachable Sets and Lyapunov-like Certificates**, *j.w. with S. Tonetta*, ATVA25, 2025.
Springer Extended version
- 9 **VeriLHyS: a Framework for LTL Specification and Verification of Hybrid Systems**, *j.w. with S. Tonetta and G. Zampedri*, TACAS26, 2026.
Springer Extended version
- 10 **SpaceEx Hybrid Models with LTL Properties**, *j.w. with S. Tonetta and G. Zampedri*, ARCH26, 2026.
Link

Talks

- Sep '20 **Seminar of Geometry group, Pisa**, *A Hyperbolic 4-manifold with a perfect circle valued Morse function.*
- Apr '22 **Seminario di Algebra e Geometria, Bologna**, *Hyperbolic 4-manifolds, perfect circle valued Morse functions and infinitesimal rigidity.*
- Oct '22 **Geometry seminars, Neuchâtel**, *Dodecahedral L-spaces and Hyperbolic 4-manifolds.*
- Nov '22 **Geometry seminars, Luxembourg**, *Hyperbolic 4-manifolds, perfect circle-valued Morse functions and infinitesimal rigidity.*
- Jul '24 **Analysis and Design of Hybrid Systems, Boulder, Colorado, Stati Uniti**, *Formal Verification of Stability for Parametric affine Switched Systems.*
- Oct '25 **Automated Technology for Verification and Analysis, Bangalore, India**, *Deriving Liveness Properties of Hybrid Systems from Reachable Sets and Lyapunov-like Certificates.*
- April '26 **Topology Seminars, Bonn**, *Can you hear the shape of a Hyperbolic Marimba?*, [Link](#).
- May '26 **Oberseminar Geometric Group Theory, Bonn**, *Can you hear the shape of a Hyperbolic Marimba?*, [Link](#).
- May '26 **Seminari del Dipartimento di Matematica, Bologna**, *Can you hear the shape of a Hyperbolic Marimba? Concerto di Superfici Iperboliche*, [Link](#).
- June '26 **Reading group on spectral geometry, Bonn**, *Sunada's construction for isospectral surfaces*, [Link](#).
- More**, I have took part in many informal seminars both online (especially during COVID pandemic, GT GAPS, Exotic LS) and offline (Pisa, Bologna, Nice, Ventotene).

Organized activities

- Mar '23 **Non-positive curvature in manifolds and groups**, *Math Day funded by INdAM*.
[Link](#)
- Mar '23 **Manifolds and groups in Bologna**, *Workshop funded by GNSAGA (INdAM)*.
[Link](#)
- '22 **TGV: Seminari di Topologia e Geometria delle Varietà**, *Department of Mathematics, Bologna*, with Stefano Francaviglia, Marco Moraschini e Stefano Riolo, We organized a weekly meeting with master students to introduce them to some research topics in differential topology.
- '18-'21 **BabyGeometri**, *Department of Mathematics, Pisa*, Weekly meetings for early-career geometers in Pisa.
[Link](#)

Referee activity

Reviewer, I have served as (sub)reviewer for several international conferences as FM, TACAS, SAFECOMP.

Scholarships

- '16-'18 **Scholarship for Mathematics Master students**, *Istituto Nazionale di Alta Matematica*.
I ranked first in the national test for this scholarship. It consisted in a written exam with several problems about Analysis, Probability, Geometry and Algebra.
- '13-'16 **Scholarship for Mathematics students**, *Istituto Nazionale di Alta Matematica*.
I won this scholarship for academic achievement, and I succeeded in renewing it for the whole duration of my Bachelor's Degree.

Visiting

- Nov '22 **Research visit**, *University of Luxembourg*.
During my visit I gave a talk in the framework of the research seminars of the geometry group, and I collaborated with prof. Jean-Marc Schlenker and his research team.
- Oct '22 **Research visit**, *University of Neuchâtel*.
During my visit I gave a talk in the framework of the research seminars of the geometry group, and I collaborated with prof. Aleksandr Kolpakov and his research team.

Teaching experience

- '21 - '22 **Support to teaching for the course *Differential Geometry*** - Department of Physics, Pisa.
- '21 - '22 **Support to teaching for the course *Geometria e algebra lineare - modulo Geometria*** - Department of Civil and Industrial Engineering, Pisa.
- '19 **Support to teaching for the course *Principles of Geometry*** - Department of Mathematics, Pisa.
- '18 **Semestral course in Arithmetic to reduce early university leaving** - Department of Mathematics, Pisa.
- '17 - '18 **Part-time Counseling (Counseling)** - Department of Mathematics, Pisa.
- '16 - '17 **Part-time Tutoring (Tutorato alla Pari)** - Department of Mathematics, Pisa.

Schools and Conferences

- Jun '26 **Conference**, *Geometry and Topology in Low Dimensions: Lasting Trends and Emerging Directions*, Budapest, Hungary, Poster presented.
- Apr '26 **Conference**, *Manifolds and groups in Bologna, IV*, Bologna, Italy, .
- Oct '25 **Conference**, *International Symposium on Automated Technology for Verification and Analysis*, Bangalore, India, Paper published.
- Sep '25 **Conference**, *Higher Dimensional Hyperbolic Geometry*, Ventotene, Italy.
- Jul '24 **Conference**, *The 8th IFAC Conference on Analysis and Design of Hybrid Systems*, Boulder, Colorado, Paper published.
- Sep '23 **Conference**, *GRAZP: Groups and Rigidity Around the Zimmer Program*.
- Jul '23 **Workshop**, *VERDI*, Porto, Portugal.
- '19-'22 **More**, *I have taken part in more than 20 international conferences on Geometry and Topology*.

Skills

Language skills

Italian	Mother tongue
English	Advanced, C1 (Cambridge certification 2024)
French	Basic, A2

Programming skills

Python	Advanced
MATLAB	Advanced
$\text{\LaTeX}$	Advanced
Sage	Intermediate
C Language	Basic

Code samples

- VeriLHyS **Tool**, I am the main contributor of the tool VeriLHyS that can be used to verify LTL properties on Hybrid Systems. Mainly written in Python and MATLAB. Available at <https://es-static.fbk.eu/tools/verilhys/>, This tool make use of several external packages and softwares (SMT solvers like cvc5 and z3, Reachable Set Computation by CORA in a MATLAB environment, Numerical Optimization by Mosek, Model Checking by nuXmv); I consider this to be an important ability to work in the area of computational mathematics..
- Paper 2 The results presented in Paper 2 were obtained using some code provided alongside the publication. This consists mainly of Sage and MATLAB scripts.
- Paper 3 The results presented in Paper 3 were obtained using some code provided alongside the publication. This consists mainly of Sage scripts. Written in collaboration with Leonardo Ferrari.
- Paper 7 Paper 7 underwent an experimental evaluation conducted with the code provided alongside the publication. This consists mainly of Python scripts. A repeatability evaluation was available, and the paper obtained the corresponding badge.
- Paper 8 Paper 8 underwent an experimental evaluation conducted with the code provided alongside the publication. This consists of a large software written in Python and MATLAB, which makes use of several software and packages like PySMT, z3, cvc5, CORA, and nuXmv. A repeatability evaluation was available, and the paper obtained the Available and Functional badges.
- Paper 9 Paper 9 is a tool paper, and as such it underwent an experimental evaluation conducted with the code provided alongside the publication. A repeatability evaluation was available, and the paper obtained the Available, Functional, and Reusable badges.
- Preprint 1 We have provided an accompanying website to the paper, available here. The repository for the full code is available here.